How Scientists Work

Scientists often work in steps like these. It is fine to go back and try again.

Name ______________________ Date ______________

1 Ask
What do I want to find out?

2 Predict
What do I think will happen?

3 Test
Try it carefully and safely.

4 Observe
Look closely. Write or draw what happens.

5 Explain
Tell someone what I found out.

The Scientific Method

This is a useful sequence, not a fixed recipe. Real investigations loop: results send you back to the question, the design or the hypothesis, and much of science does not involve an experiment at all.

Name ______________________ Date ______________

1 **Ask a question** Specific, measurable and testable.

2 **Background research** What is already known? Note your sources.

3 **Form a hypothesis** A testable statement that could be shown wrong.

4 **Design the investigation** Name the independent, dependent and controlled variables.

5 **Collect data** Repeat trials. Record units and uncertainty.

6 **Analyze the data** Tables, graphs, averages. Look for patterns.

7 **Draw a conclusion** Do the data support the hypothesis, or not? Either result is a real result.

8 **Communicate** Report what you did and what you found; ask the next question.

The Scientific Method

Scientists often follow steps like these. Real investigations rarely run straight through - you may loop back, change something and test again.

Name ______________________ Date ______________

1. **Ask a question** Something you can actually test.

2. **Find out what is known** Books, websites, a teacher.

3. **Write a hypothesis** A testable idea, not just a guess.

4. **Plan a fair test** Change one thing; keep the rest the same.

5. **Observe and record** Measure carefully and write it down.

6. **Look for patterns** What does the data show?

7. **Conclude and share** Answer the question. Say what you would do differently.